

The influence of the bullseye versus standard fixation cross on eye movements and classifying natural images from EEG

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1 Abstract

Eye movements are a ubiquitous and natural behavior, but in many tightly controlled experimental visual paradigms eye movements are undesirable. Their occurrence can pose challenges to the interpretation of behavioral and neuroscientific data, in particular for magneto- and electroencephalography (M/EEG) which is sensitive to signals created by eye muscle movement. Here we compared the effect of two different fixation symbols – the standard fixation cross and the bullseye fixation cross – in the context of a visual paradigm with centrally presented naturalistic object images. We investigated eye movements and EEG data recorded simultaneously using behavioral and multivariate analysis techniques. Our findings comparing the bullseye to the standard fixation cross are threefold. First, the bullseye fixation cross reduces the number of saccades and amplitude size of microsaccades. Second, the bullseye subtly reduces classification accuracy in both eye tracking and EEG data for the classification of single object images, but not for the superlevel category animacy. Third, using representational similarity analysis we found a systematic relationship between eye tracking and EEG data at the level of single images for the standard, but not for the bullseye fixation cross. In conclusion, we recommend the bullseye fixation cross in experimental paradigms with fixation when particularly tight control of fixation is beneficial.

2 Introduction

Eye movements are a diverse, ubiquitous, and integral part of visual behavior¹. For example, we use saccades, i.e. large voluntary eye movements, to explore a scene, and microsaccades, i.e. small involuntary eye movements, to keep the retinal image from fading².

However, for human cognitive neuroscience experiments that aim to establish statistical dependencies between tightly-controlled visual input and brain activity eye movements pose experimental challenges. Eye movements change the visual input to the brain, and also influence the recordings of brain activity by magneto- and electroencephalography (M/EEG)^{3,4} through the currents created by eye muscle movements. Eye movements can thus introduce noise, add confounds, or both simultaneously into the experimental settings as shown for analyses of event-related potentials^{5,6}, frequency-resolved responses⁷, and multivariate activation patterns^{8–10}.

A straight-forward way to reduce the effect of eye movements is to avoid them in the first place and ask participants to fixate. However, participants do not follow such instructions perfectly - novice participants in particular do not control their eye movements accurately¹¹, and fixation behavior varies widely among participants^{11–13}.

The amount of residual eye movements depends on the type of visual symbol used as a fixation target. An influential study¹⁴ systematically evaluated the effect of different fixation symbols on eye movements when presented on a uniform gray background, finding that a so-called bullseye fixation cross was associated with the smallest number of eye movements. If this observation generalized to situations in which the fixation symbol appeared on top of visual stimuli for which brain responses were recorded, it should reduce the effect of eye movements on the neural analysis.

To investigate we collected eye tracking and EEG data simultaneously while participants viewed naturalistic still images of everyday objects overlaid with one of two different fixation symbols: a classical fixation cross and the bullseye fixation cross. We used time-resolved multivariate analysis methods to establish statistical dependencies between the experimental stimuli and the EEG data.

As anticipated, we replicate the behavioral advantage of the bullseye fixation cross over a standard fixation cross for experimental setups involving naturalistic still images¹⁴. We further show that the bullseye fixation cross reduced the influence of eye movements on the analysis of EEG brain data by reducing stimulus-specific eye movements. We thus recommend using the bullseye fixation cross to avoid unwanted eye movements and their effects on brain measurements in studies involving static, naturalistic still images.

3 Materials and methods

3.1 Participants

30 healthy adults (20 female, 9 male, 1 diverse, mean age = 24.77, SD = 4.08) were recruited at Freie Universität Berlin. Participants provided informed consent and received either course credit or monetary reimbursement for their participation. All participants had normal vision and no history of neurological disorders. The experiment was approved by the Ethics committee of Freie Universität Berlin and was conducted in accordance with the Declaration of Helsinki.

3.2 Stimulus set and stimulus presentation

The main stimulus set consisted of 40 images of everyday objects on natural backgrounds (Fig. 1a). 20 images depicted animate, and 20 inanimate objects. The set is a subset of the stimulus set used in a previous study¹⁵.

Stimuli subtending 5-degree visual angle were presented centrally on a gray background on a Samsung Screen (SyncMaster 2233) (Fig. 1b) and either overlaid with a standard or bullseye fixation cross subtending 0.6-degrees visual angle (Fig. 1c).

Stimulus presentation was controlled by custom-made scripts in MATLAB 2021a¹⁶ and the Psychtoolbox extension^{17–19}.

3.3 Experimental design

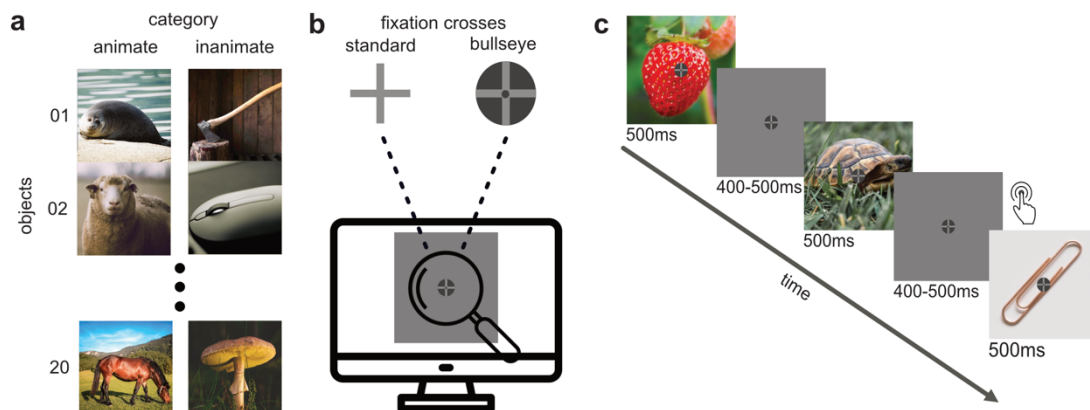


Fig.1: Stimulus set and paradigm. **a)** The stimulus set consisted of 40 images with natural objects, split equally into the animate and inanimate subcategories. **b)** We used two different fixation crosses overlaid onto the stimuli: the standard fixation cross and the bullseye fixation cross. **c)** Trials were blocked by fixation symbol. Participants were instructed to fixate the fixation symbol while objects appeared for 500 ms in random order with an inter-stimulus-interval of 400-500 ms and not to blink. Every 4-6 trials the image of a paperclip was shown for which participants were asked to press a button and to blink their eyes. Due to copyright, the stimuli in this figure are not the exact stimuli used in the experiment but resemble the original.

The combined EEG and eye tracking study consisted of one session, partitioned into 14 blocks. In each block only one fixation symbol was used, resulting in 7 blocks for the fixation cross and 7 blocks for the bullseye fixation cross. The order of blocks was randomized.

During each block, stimuli were presented with object images in random order. On each trial, an image overlaid with a fixation symbol was presented for 500 ms, followed by an inter-trial interval randomly varying between 400 and 500 ms.

Participants were instructed to fixate on the fixation symbol in the middle of the screen and not to blink their eyes. Every 4th-6th trial an image of a paper clip was shown that was not part of the main stimulus set. Participants were instructed to press a button, blink, and refrain from blinking otherwise. Responses were collected with a standard keyboard, attached to the presentation computer. Participants could take self-paced breaks after each block. On average participants detected 81,9% of catch trials.

In each block each image of the main stimulus set was repeated six times, resulting in 240 trials plus 60 additional paperclip trials. Over the course of the experimental session, this resulted in 3380 trials for the main stimulus set and 840 paperclip target trials. Paperclip trials were excluded from further analysis

3.4 Eye tracking recording and preprocessing

3.4.1 Recording

We monocularly recorded the right eye of each participant using the Eyelink 1000 Tower Mount (SR Research Ltd., Osgoode, Ontario, Canada) and the Eyelink Toolbox extensions²⁰ with a sampling rate of 1000 Hz. Participants were seated comfortably in front of the screen and instructed to rest their chins on the chinrest, 60cm away from the monitor. Before each block, we calibrated the eye tracker using a nine-point calibration. The data was recorded in gaze position coordinates.

3.4.2 General preprocessing

Before applying the saccade detection algorithm, we cleaned the eye tracking data from artifacts by excluding (i) all trials containing blinks²¹, (ii) all data samples outside the screen range, (iii) all negative data samples, and (iv) additionally all data samples in a 100 ms period around excluded samples from steps one to three²².

After artifact exclusion we converted the eye tracking data samples from screen coordinates in pixels to spherical angles in degree, using the following formula:

$$\beta_x = 2 * \text{atan2}(p_x * m, d)$$

where β_x is the azimuth angle in visual degree from the monitor center, p_x is the horizontal position relative to the center of the monitor in pixels, m is the unit necessary to convert the pixel of the monitor to mm and d is the distance to the monitor in mm²¹. An equivalent procedure was applied to the y-coordinate. This yielded a data frame with x and y coordinates for the position of the eye in degree visual angle (dva) for each time point for all included trials.

3.4.3 Saccade and microsaccade detection

We used a velocity-based algorithm²³ to detect saccades and microsaccades. Numerous studies have shown that the magnitude of microsaccades mostly falls below 1 degree^{23–26}. Therefore, we defined all saccades with amplitudes smaller than one degree as microsaccades and all saccades with amplitudes larger or equal to one degree as saccades.

To detect saccades and microsaccades, the position vector (x and y positions of the eye in dva) was transformed into a two-dimensional velocity space. The velocity of the eye was required to exceed eight standard deviations of the eye's velocity during the trial for at least 8 ms to be detected as an event. These values are higher than the typically proposed values²³ to minimize noise, emerging from our monocular recording setup (R. Engbert, personal communication).

To further boost the signal-to-noise ratio, we implemented a ratio criterion, where data points were only considered microsaccades if the ratio of path length to amplitude exceeded 0.5¹⁴. Together the procedures yielded a data frame with all detected saccades and microsaccades and their amplitudes for all trials.

3.4.4 Preprocessing for multivariate pattern analysis

We took specific steps to preprocess the eye tracking data for multivariate pattern analysis (MVPA). We epoched the data from -200 to +1000 ms around stimulus onset, downsampled the data to 200 Hz, and baseline corrected each epoch by subtracting the mean of the 200 ms prestimulus interval from the entire epoch. Each epoch thus contained a time course of x and y positions of the eye.

We only kept trials for MVPA that were neither excluded during the eye tracking data preprocessing, nor the EEG data preprocessing (see below). This amounted to on average 40 trials per object image and 802 trials per category (animate/inanimate) per participant.

3.5 EEG recording and preprocessing

3.5.1 Recording

We recorded EEG data using the ActiCap64 electrodes system and Brainvision actiChamp amplifier. 64 Electrodes were placed according to the 10-10 system²⁷ with an additional ground and reference electrode placed on the scalp. We recorded the data with Brainvision recorder software, using a 1000 Hz sampling rate and online filtering between 0.03 Hz and 100 Hz²⁸. We kept all impedances below 10kΩ.

3.5.2 Preprocessing

We preprocessed the EEG data offline using Fieldtrip²⁹ in MATLAB 2021a¹⁶. We epoched the data between -200 and +1000 ms relative to stimulus onset, notch filtered it at 50 Hz, downsampled it to 200 Hz, and performed baseline correction by subtracting the mean of the 200 ms prestimulus interval from the entire epoch. Each epoch thus contained a 64-dimensional EEG time course. Subsequently, we excluded all trials which were excluded during the eye tracking preprocessing (see above for details) from the EEG data, and additionally manually removed all channels and trials containing excessive noise. We then interpolated missing channels by using the average of all surrounding channel neighbors. This procedure resulted in the equivalent number of matched EEG and eye tracking trials, i.e., on average 40 trials per object image and 802 trials per category (animate/inanimate) per participant.

3.6 Generalized linear mixed model

We analyzed saccade and microsaccade numbers and amplitudes using Generalized Linear Mixed Models (GLMMs), with the lmer4 package³⁰ in R version 1.3.1093³¹. Saccades and microsaccades were analyzed separately. A general description of the models is given by the following formula:

$$\begin{aligned} g(model_{baseline}) &= \beta_0 + u_{0,j} + e_0 \\ g(model_{fixation}) &= \beta_0 + \beta_1 * cross + u_{0,j} + u_{1,j} * cross + e_0 \end{aligned}$$

with $g()$ defining the link function, β_0 the intercepts, $\beta_1 * cross$ the fixed effect fixation cross, $u_{0,j}$ the subjects' random intercept, $u_{1,j} * cross$ the random slopes for the factor fixation cross and e_0 the error term. We included random intercepts to capture variances in the individual subject means, and random slopes to allow for participant-specific effect magnitudes³².

We modeled the effect of the factor fixation cross on amplitude size and the number of saccades and microsaccades separately. In each case, we fitted two models (baseline and full model), with a linear link function for the amplitude and a Poisson link function for the number of saccades and microsaccades. The full model included the factor fixation cross, whereas the baseline model did not. We used sum coding (-0.5 and 0.5) for all contrasts to be able to interpret differences in condition means.

3.7 Multivariate pattern analysis

To characterize the time course with which object and category representations emerge we conducted MVPA using linear support vector machine (SVM)³³ as implemented in libsvm³⁴.

We conducted 8 separate analyses: 2 (modality: EEG, eye tracking) * 2 (classification type: image identity, object animacy) * 2 (fixation cross: standards, bullseye), and each participant was analyzed separately.

Each analysis had three main steps, with each step performed independently for each time point in the epoch. First, we averaged over individual trials to create pseudo-trials, thereby increasing the signal-to-noise ratio³⁵. Second, we trained an SVM on all but one pseudo-trial to differentiate either image identity or image category in a pair-wise fashion. Third, we tested the prediction accuracy of the SVM on left-out data.

We conducted each analysis in two ways: time-resolved and time-generalized MVPA. For time-resolved analysis^{36,37}, the SVM was trained and tested on data from the same time points only, yielding a single time course with which object information emerges as a result. For time-generalized analysis³⁸, the SVM was trained and tested for all possible combinations of time points, yielding a two-dimensional result array, indicating how stable EEG activation patterns are. We describe the details of each analysis type below.

3.7.1 Time-resolved MVPA

We used time-resolved MVPA to determine the time course with which object identity and animacy representations emerged. We conducted equivalent analyses based on EEG and eye tracking data. For each time point, we extracted trial-specific EEG channel activations (64 channels, for EEG classification) or trial-specific eye positions (x and y coordinates in dva, for eye tracking classification).

We averaged trials aggregated for each object (for object classification) or by animacy (for classification animate vs. inanimate) into 6 pseudo trials³⁵. We used multivariate noise normalization³⁹ to whiten the data and further improve the signal-to-noise ratio⁴⁰. For this, we multiplied the data by the inverse of the square root of the covariance matrix of electrode activations from the entire epoch. We trained the SVM classifier in a pair-wise fashion on data of all but one pseudo trial and tested the SVM classifier on the left-out trial. We repeated this procedure 100 times, each time with a different random assignment of trials to pseudo-trials. To ensure that the SVM classifier was not biased by an excess of data for one or the other condition we included the same number of trials in the creation of pseudo-trials for each condition. We averaged the resulting decoding accuracies over the 100 repetitions.

Across our analysis space, this resulted in one decoding accuracy number per participant, modality, classification type, and fixation cross and time point.

3.7.2 Time generalization MVPA

We used time generalization MVPA³⁸ to determine the temporal stability of object and animacy representations. The procedure was equivalent to the time-resolved MVPA described above, except that we tested the SVM classifier trained on any one specific time point iteratively on all time points. Further, for object classification, the EEG and eye tracking data were downsampled to 50 Hz for time and memory efficiency.

Across our analysis space, this resulted in a 2D matrix of classification accuracies, indexed in rows and columns by the time points of the epoch, per participant, modality, classification type, and fixation cross.

3.8 Representational similarity analysis

We performed representational similarity analysis (RSA)⁴¹ to quantitatively compare data across modalities (EEG, eye tracking) This was a two-step process. First, for each participant,

we constructed representational dissimilarity matrices (RDMs) for each modality and fixation cross type separately. Second, we related the RDMs directly by calculating their similarity. We describe each of the two steps in detail below.

3.8.1 Construction of RDMs

We created RDMs using the preprocessed EEG and eye tracking data for each fixation cross type separately. For each time point, we calculated dissimilarity for all pairs of conditions (i.e., the 40 different object images) by using 1-Pearson correlations³⁹ (Fig. 5a). This resulted for each time point in a 40×40 RDM, indexed in rows and columns by the conditions compared. We equalized the number of trials across conditions by randomly subsampling trials with the minimum number of trials across conditions. We repeated the analysis 100 times, calculating an RDM each time, and averaged the RDMs across the iterations. We used the vectorized upper triangular matrix of the symmetric RDM (without the diagonal) for further analysis.

3.8.2 RDM comparison

We determined how visual representations measured with EEG and eye movements measured with eye tracking relate. For this, we correlated the EEG RDMs and eye tracking RDMs using Spearman's r for each participant, time point, and fixation cross separately. Averaging over participants we obtained a time course of similarity between the EEG and eye tracking data for each fixation cross type.

3.8.3 Noise ceiling

The RSA results are bound by measurement noise as the calculated correlation values are affected by both the variance in the data as well as by noise in the data. To determine how much variance could in principle be explained by the model if we knew the true data-generating-model^{42,43} we determined the noise ceilings for the EEG and eye tracking data. We estimated an upper and lower bound for the noise ceiling⁴³ in a time-resolved fashion. For the lower bound, we singled out a participant RDM and compared it with the average of the RDMs for all participants except the one singled out, iterated this for each participant, and averaged. For the upper bound, we singled out a participant RDM and compared it with the average of the RDMs for all participants including the one singled out, iterated for each participant, and averaged.

3.9 Statistical testing

MVPA-related statistical analyses were performed using MATLAB. We evaluated the statistic of interest (classification accuracy, correlation coefficient) using non-parametric tests. The null hypothesis was that the statistic of interest was equal to chance (i.e., 50% classification accuracy or a Spearman's r of 0). To estimate a null distribution we used a sign permutation test, multiplying participant-specific data randomly multiplied with +1 or -1 and recomputing the statistic of interest 10,000 times. To obtain p-values we calculated the rank of the test statistic with respect to the null distribution.

We controlled the family-wise error rate across time points using cluster size inference⁴⁴ with a $p < 0.05$ cluster-definition threshold and $p < 0.05$ cluster threshold. For all tests, both thresholds were right-sided with the exception of EEG classification difference curves where the thresholds were two-sided.

We used bootstrapping to compute 95% confidence intervals for peak latencies. For this, we sampled the participant pool 10,000 times with replacement and calculated peak latencies for each sample. This created an empirical distribution of peaks on which we determined 95% confidence intervals.

For the GLMMs we performed statistics using R version 1.3.1093³¹. The derivation of p-values for the different predictors in GLMMs is debated with two prominent candidate approaches- the Wald test and the likelihood ratio test (LRT). For Wald tests, the z distribution is used to obtain p-values from Wald t-values. This is generally appealing as the t distribution approximates the z distribution for increasing numbers of degrees of freedom (identical for infinite degrees of freedom). LRTs are classically used to test whether a certain predictor should be part of a model or whether the predictor can be excluded. LRTs determine which and whether one of two models fits the data better. We report the outcome of both tests that here in all cases concur.

4 Results

We investigated the effect of fixation cross type on naturalistic still images in a three-step procedure. First, we assessed eye movement behavior alone, analyzing the number and amplitudes of saccades and microsaccades using GLMMs. Second, we in parallel assessed EEG and eye tracking data using MVPA, delineating how fixation cross type influences classification of visual information presented alongside the fixation cross. Third, to directly and quantitatively evaluate systematic relationships between eye tracking data to EEG data we used RSA.

4.1 The bullseye fixation cross reduces eye movements

We analyzed the influence of fixation cross type on eye movements as an influence on the number and amplitude of saccades. We used a velocity-based algorithm²³ to detect saccades and microsaccades, with microsaccades being smaller than 1 degree of visual angle.

Visual inspection of the descriptive statistics for number and amplitude of saccades and microsaccades (Fig 2a,b) revealed similar distributions, indicating that any potential effects of fixation crosses are subtle. To statistically evaluate the effect of fixation cross type on saccade and microsaccade amplitude and numbers we fitted pairs of GLMMs, once with the predictor fixation cross type included and once excluded. We evaluated statistical significance with both the likelihood ratio test (LRT) and the Wald test, yielding equivalent results (see Table 1 for amplitudes, Table 2 for number).

Using the LRT, we found that the standard fixation cross was associated with a 3.4% increase in saccades ($\chi^2(1) = 10.35$, $p = 0.001297$) and a 0.023dva increase in amplitudes ($\chi^2(1) = 7.96$, $p = 0.004785$). There was no evidence that fixation cross type affected saccade amplitudes and microsaccade numbers.

Together this shows that the bullseye fixation cross reduces eye movements compared to the standard fixation cross when participants are asked to fixate on naturalistic still images. Our results contextualize the effect size as subtle with respect to the overall observed distribution of eye movements and generalize previous findings from fixation on uniform backgrounds¹⁴.

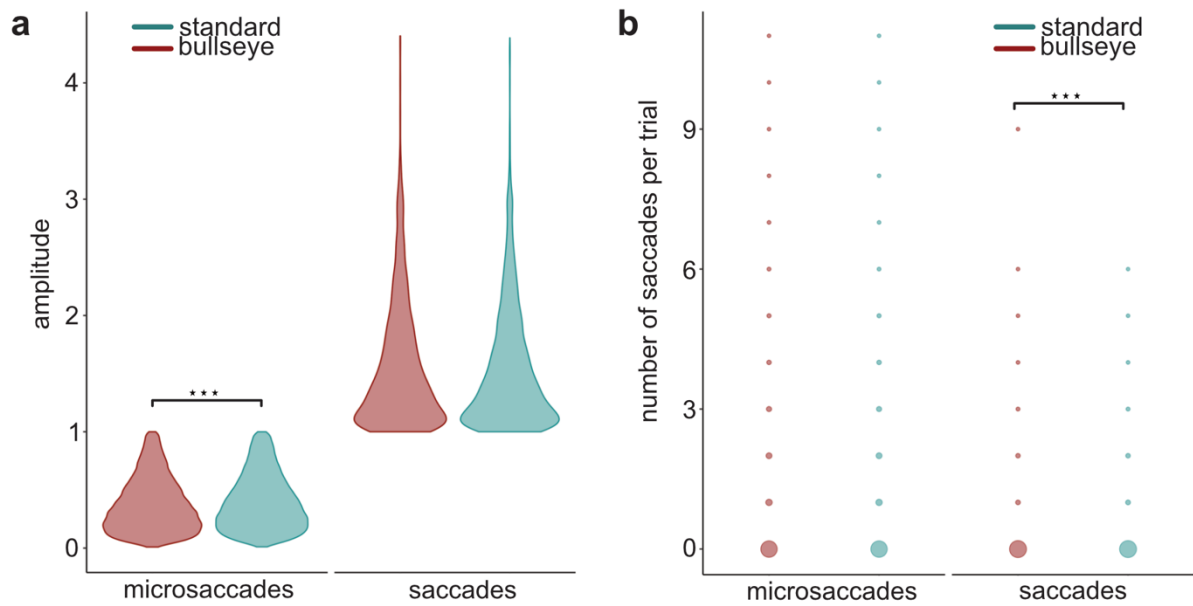


Fig. 2: Descriptive statistics for the amplitude and number of microsaccades and saccades per trial. **a)** Violin plots of saccade and microsaccade amplitudes. **b)** Distribution of number of saccades and microsaccades per trial. Stars indicate significant fixed effects with $p < 0.01$.

	saccades		microsaccades	
	(1)	(2)	(3)	(4)
intercept	1.553*** (1.485, 1.620)	1.551*** (1.483, 1.619)	0.369*** (0.342, 0.395)	0.378*** (0.351, 0.405)
fixation cross		-0.006 (-0.040, 0.027)		0.023*** (0.008, 0.037)
log likelihood	-8,805.197	-8,805.133	7,123.959	7,127.938
Akaide inf. crit.	17,620.400	17,622.270	-14,237.920	-14,243.880
Bayesian inf. crit.	17,657.220	17,666.460	-14,191.040	-14,191.040

Note: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 1: Summary of model fits, intercepts, and fixed effect estimates for saccade and microsaccade amplitude models. Estimates are in real values and represent the difference between conditions. Stars represent significance, calculated with Wald's t-as-z approach. Confidence intervals are stated in parentheses below.

	saccades		microsaccades	
	(1)	(2)	(3)	(4)
intercept	0.011*** (0.008, 0.015)	0.012*** (0.008, 0.017)	0.158*** (0.139, 0.181)	0.149*** (0.130, 0.170)
fixation cross		1.256*** (1.109, 1.421)		1.034 (0.988, 1.082)
log likelihood	-51,131.250	-51,126.070	-266,203.600	-266,202.800
Akaide inf. crit.	102,270.500	102,262.100	532,415.200	532,415.600
Bayesian inf. crit.	102,315.000	102,317.800	532,459.700	532,471.200

Note: * p<0.1; **p<0.05; ***p<0.01

Table 2: Summary of model fits, intercepts, and fixed effect estimates for saccade and microsaccade number models. Estimates are in rate ratios and represent the difference between conditions. Stars represent significance, calculated with Wald's t-as-z approach. Confidence intervals are stated in parentheses below.

4.2 The impact of fixation cross type on MVPA of visual information from eye tracking and EEG data

We investigated how fixation cross type influences the classification of visual information available in the naturalistic object images on which the fixation cross symbols were overlaid. For this, we analyzed eye tracking and EEG data in parallel to identify common patterns that might suggest an influence of eye movements on EEG data. We assessed visual information at two levels commonly queried in current cognitive neuroscience experiments: at the level of single object images, and at the higher categorization level by classifying object animacy (i.e., animate vs. inanimate). We present the classification results for the eye tracking data first.

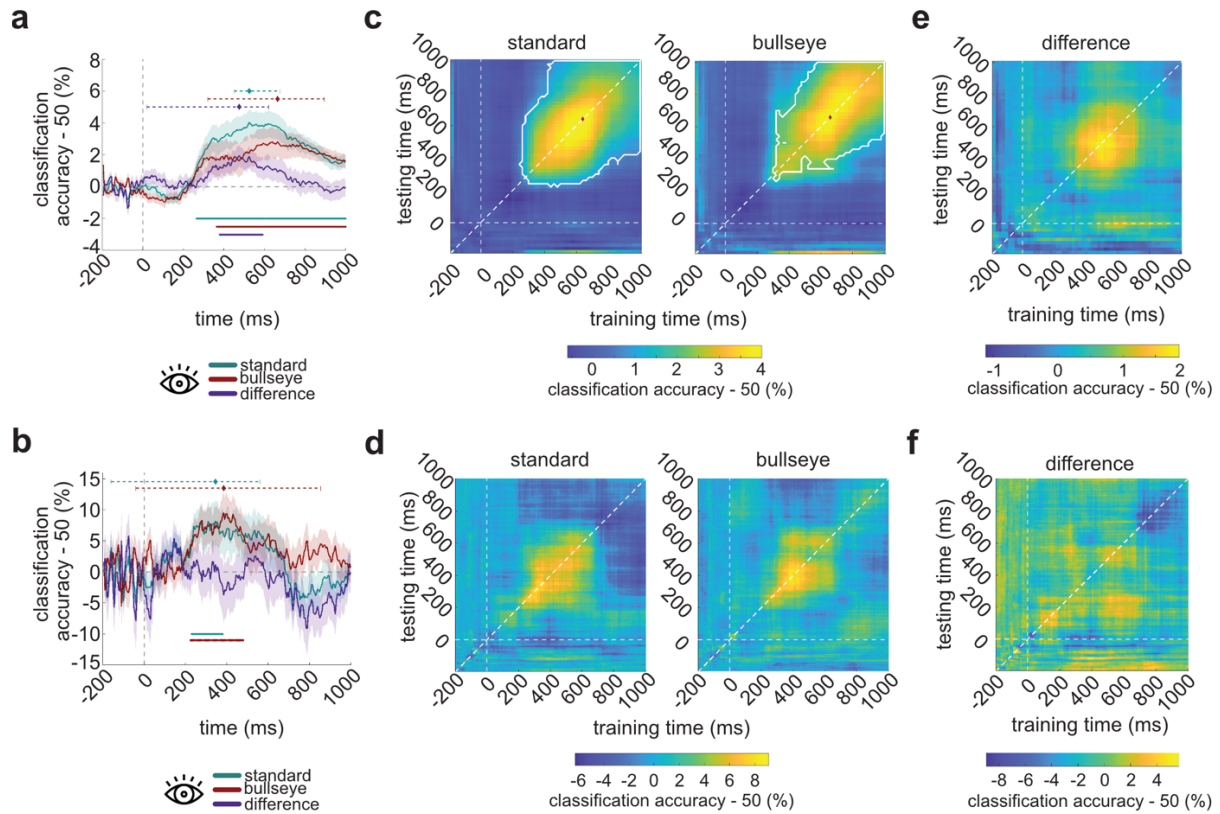


Fig. 3: Eye tracking MVPA results. **a)** Time-resolved object and **b)** animacy classification. The vertical dotted line shows stimulus onset, error bars indicate SEMs across participants. Peak latencies and 95% confidence intervals are indicated above the curves as dotted lines color coded as the result curves. The lines below the curves denote significant time points. **c)** Object and **d)** animacy time-generalization analyses, and **e,f)** differences between fixation crosses for each. Statistically significant time points are outlined in white. The diamond shapes indicate peak latencies. The dashed lines indicate stimulus onset and the diagonal. Results are corrected for multiple comparisons by cluster correction (cluster definition threshold $p < 0.05$, cluster threshold $p < 0.05$). Detailed information about cluster extents can be found in Table 3.

We determined whether there is a systematic relationship between eye movements and the visual material presented. For this we applied time-resolved multivariate pattern analysis to the eye tracking data classifying both object/image identity (Fig. 3a) and object animacy (Fig. 3b). We conducted the classification analyses separately for the two fixation cross types and compared the outcome. We assessed statistical significance with cluster permutation tests (cluster-definition threshold $p < 0.05$; cluster threshold $p < 0.05$) and reported peak latencies with 95% confidence intervals in square brackets.

For object identity classification we found significant information for both fixation crosses (Fig. 3a, turquoise and red curve). The result curves increased gradually from 200 ms after stimulus onset, followed by a prolonged plateau and slow decay over the duration of the trial with peaks at 525 ms [450 675] and 665 ms [320 895]. The difference curve between those results (Fig. 3a, purple curve) had a similar shape, showing higher accuracy for the standard than the bullseye fixation cross with a peak at 475 ms [20 620].

For animacy classification, we also found significant information for both fixation crosses (Fig. 3b, turquoise and red curve) with peaks at 345 ms [-160 560] and 385 ms [-40 855] respectively. However, the difference curve fluctuated around chance level and was not significantly different from it.

Together, these results indicate a systematic relationship between eye movements and the visual material presented at the level of single images and image identity, but not at the level of more abstract, categorical object divisions.

classification scheme	fixation cross type		
	standard	bullseye	difference
object identity	265-1000 ms	365-1000 ms	380-590 ms
animacy	230-1000 ms	225-1000 ms	no significant clusters

Table 3: Cluster extent of time-resolved eye tracking MVPA. Overview of earliest and latest time points of significant clusters for the different fixation cross types and classification schemes.

4.2.2. Time-generalized visual information classification on eye tracking data

The finding of prolonged object information in the time-resolved eye tracking classification analysis poses the question about the temporal stability of the data patterns underlying this result: is the prolonged effect due to a data pattern stable over time, or due to a rapidly evolving data pattern? To assess temporal stability we conducted time-generalized MVPA³⁸, classifying object information across time points in the epoch.

Visual inspection of the results suggested a stable data pattern as classification generalized across data points for both object identity (Fig. 3c) and object animacy (Fig. 3d), and for both fixation crosses. However, statistically results were significant only for object identity.

As expected from the time-resolved analysis, visually we observed higher classification accuracy for the standard than the bullseye fixation cross (Fig. 3e), with strong off-diagonal classification results again indicating temporal stability of the underlying data patterns. However, neither the difference curve for object identity nor for object animacy (Fig. 3e,f) was statistically significant, precluding interpretation.

Together this shows that object identity classification from eye tracking data depends on a temporally stable rather than strongly dynamic data patterns.

4.2.3 Classifying visual information from EEG data

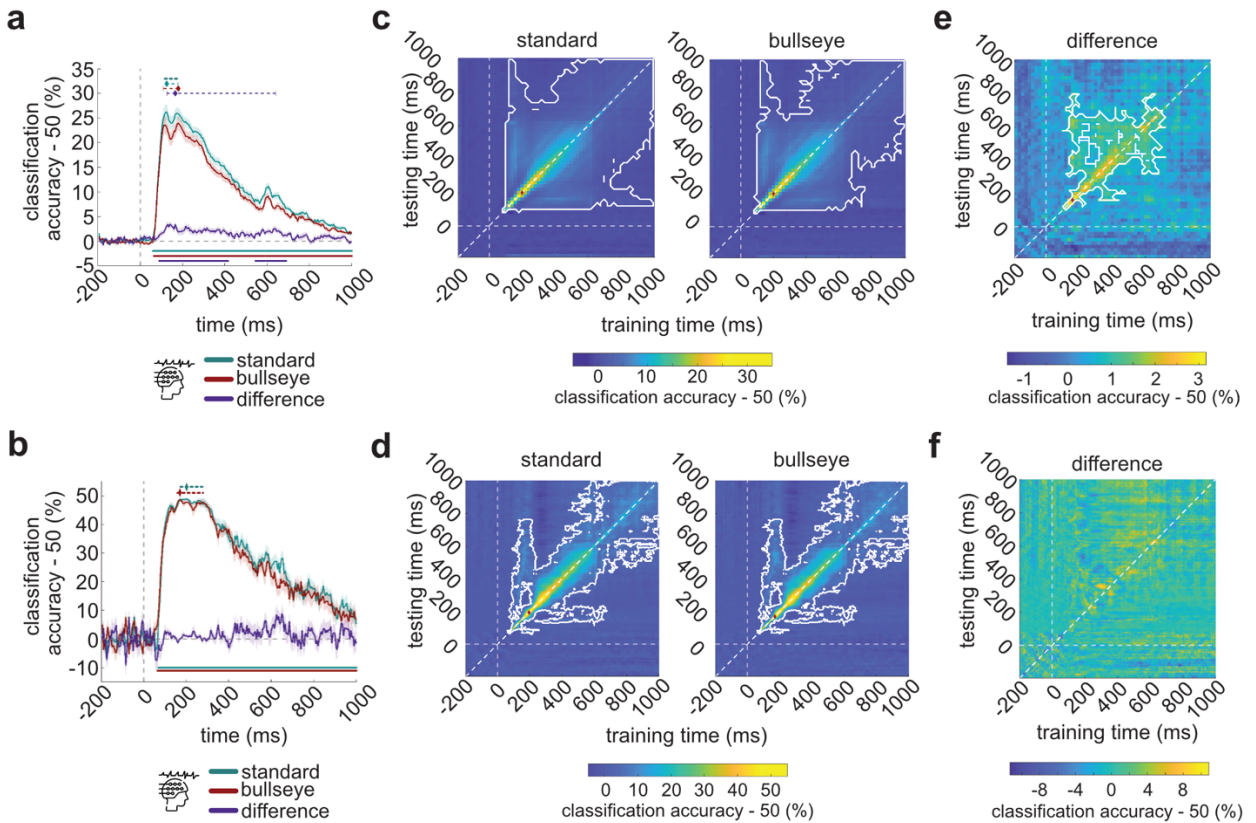


Fig. 4: EEG MVPA results. **a)** Time-resolved object and **b)** animacy classification. The vertical dotted line shows stimulus onset, error bars indicate SEMs across participants. Peak latencies and 95% confidence intervals are indicated above the curves as dotted lines color coded as the result curves. The lines below the curves denote significant time points. **c)** Object and **d)** animacy time-generalization analyses, and **e,f)** differences between fixation crosses for each. Statistically significant time points are outlined in white. The diamond shapes indicate peak latencies. The dashed lines indicate stimulus onset and the diagonal. Results are corrected for multiple comparisons by cluster correction (cluster definition threshold $p < 0.05$, cluster threshold $p < 0.05$). Detailed information about cluster extents can be found in Table 4.

Using an equivalent analysis strategy as for the eye tracking data, we determined whether there is a systematic relationship between EEG data and the visual material presented, and to which degree this is influenced by fixation cross type

For object identity classification we found significant information for both fixation crosses (Fig. 4a, turquoise and red curve). The result curves increased rapidly from 60 ms after stimulus onset, followed by peaks at 125 ms [120 180] (standard fixation cross) and at 180 ms [110 185] (bullseye fixation cross), and a gradual decline. This result mirrors the commonly observed MVPA results pattern for classifying visual information at the image level from M/EEG^{45–49}. The difference curve between results of classification for the different fixation crosses (Fig. 4a, purple curve) had a similar, though highly down-scaled shape with a peak at 165 ms [125 640] with higher accuracy for the standard fixation cross.

For animacy classification, we also found significant information for both fixation crosses (Fig. 4b, turquoise and red curve) with peaks at 200 ms [170 280] (standard fixation cross) and 170 ms [160 280] (bullseye fixation cross), consistent with previous results^{45–47,50}. However, the difference curve (Fig. 4b, purple curve) fluctuated around chance level and was not significantly different from it.

In sum, these results indicate an influence of fixation symbols on classification of visual

information from EEG at the level of single images, but not more abstract, categorical object divisions. This results pattern parallels that from eye tracking classification, suggesting that observations of stronger eye movements for the standard fixation cross compared to the bullseye fixation cross are related to the increased classification accuracy in the EEG data analysis.

classification scheme	fixation cross type		
	standard	bullseye	difference
object identity	60-1000 ms	65-1000 ms	90-415 ms, 545-715 ms
animacy	70-1000 ms	65-1000 ms	no significant clusters

Table 4: Cluster extent of time-resolved EEG MVPA. Overview of earliest and latest time points of significant clusters for the different fixation cross types and classification schemes.

4.2.4 Time-generalized visual information classification from EEG data

We assessed temporal stability of EEG activation patterns underlying time-resolved EEG classification with the same methodology as for eye tracking data patterns using time generalization analysis.

For both object identity and object animacy we observed evidence for both rapidly changing as well as more stable activation pattern dynamics for both fixation cross types (Fig. 4c,d), consistent with previous studies investigating perception with time-generalized analysis^{8,45,46,51}. The rapidly changing dynamics were indicated by the relatively high classification accuracy along the diagonals, and the stable aspects by significant effects far beyond the diagonal, in particular a broadening of effects after 200 ms.

The difference curve for object identity classification (Fig. 4e) comparing results based on fixation cross symbols revealed both rapidly changing and stable activation pattern dynamics underlying the higher classification accuracy for the standard compared to the bullseye fixation cross. The difference curve for animacy classification (Fig. 4f), as predicted from the time-resolved analysis (Fig. 4b), was not significant.

Together these results show that fixation cross type influences EEG classification in both rapidly changing as well as stable neural dynamics, with higher classification for the standard fixation cross.

4.3 The bullseye fixation cross reduces systematic eye movement-related confounds in EEG classification

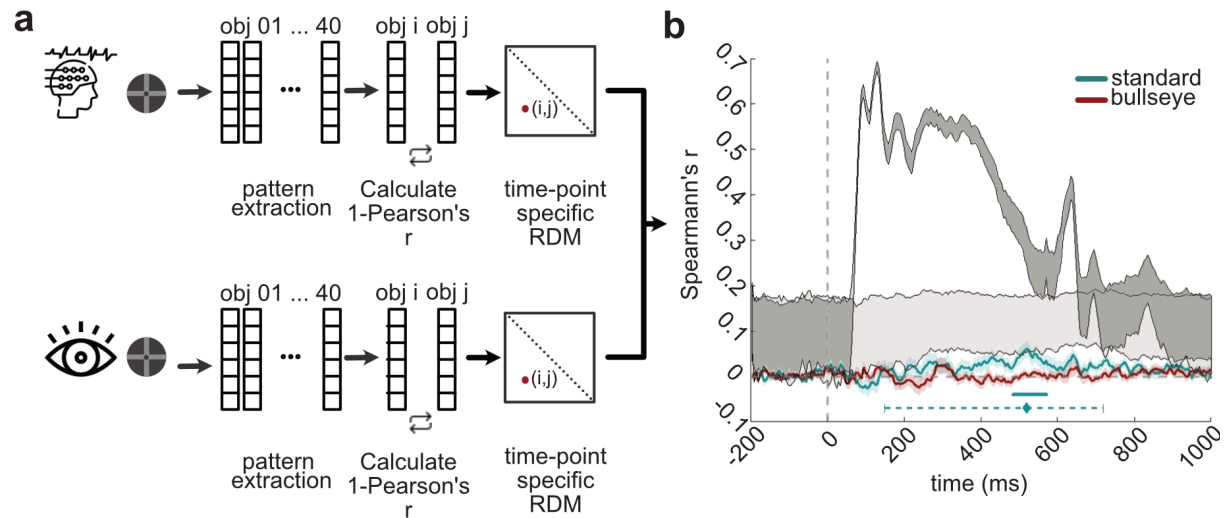


Fig. 5: Relating EEG and eye tracking data using RSA. **a)** Analysis pipeline for time-resolved RSA. In a first step, we built RDMs separately for EEG and eye tracking data, and for both fixation crosses. For this we calculated the pairwise dissimilarity ($1 - \text{Pearson's } r$) between all 40 individual object conditions, resulting in 40×40 RDMs for each time point. In a second step we compared RDMs using Spearman's r . **b)** RSA results. The vertical dotted line shows stimulus onset, error bars indicate SEMs across participants. The lines below the curves denote significant time points. Peak latencies and the corresponding CIs are indicated by the dotted line below the curve. The dark and light grey shaded areas delineate the upper and lower bounds on the noise ceilings for the EEG and the eye tracking data respectively.

Our analyses revealed - both for eye movement and for EEG data - a decrease of classification accuracy for object identity classification for the bullseye fixation cross compared to the standard fixation cross. This suggests the hypothesis that the decrease in EEG classification accuracy is systematically related to a decrease in eye movement-related effects affecting the EEG, such as confounding eye muscle activity or differences in neural processing as a consequence of eye movements.

A prediction of this hypothesis would be that images eliciting similar activation patterns in the EEG data should also elicit similar activation patterns in the eye tracking data. To test this prediction directly and quantitatively we used representation similarity analysis^{41,52,53}. In a time-resolved fashion, and for each fixation symbol separately, we aggregated all pairwise dissimilarities between object identities in representational dissimilarity matrices that abstract away from the disparate measurement spaces of eye tracking and EEG into a common similarity space (Fig 5a). We then related the eye tracking and EEG data by calculating the similarity between their RDMs.

This analysis revealed a systematic relationship between EEG and eye tracking data for the bullseye fixation cross, but not for the standard fixation cross (Fig. 5b). This shows that for the standard fixation cross condition eye movements confound EEG data to a small but systematic degree. In contrast, the difference curve between the RSA results for the two fixation symbols was not significant.

5 Discussion

5.1 Summary

We compared the effect of two fixation cross types - the standard and the bullseye fixation cross - on eye movements and EEG data in the context of a paradigm presenting naturalistic object images that were either animate or inanimate. We made three key observations. First, we showed that the bullseye fixation cross reduced eye movements compared to the standard fixation cross. Second, we showed that the bullseye fixation cross reduced classification of object identity, but not animacy from both eye tracking and EEG data. Third, we established a systematic relationship between classification results from eye tracking and from EEG data for the standard fixation cross, but not the bullseye fixation cross.

5.2 The bullseye fixation cross reduces eye movements

Previous research established that the bullseye fixation cross reduces eye movements compared to the standard fixation cross (and other fixation crosses) when presented in isolation on a uniform gray background¹⁴. Here we extend this research by reproducing the advantage of the bullseye fixation cross when superimposed on naturalistic object images. This result generalizes the previous findings¹⁴ to a commonly used basic visual paradigm in cognitive neuroscience. This is further not a trivial result, as the naturalistic images were neither controlled for color, luminance nor salience, while saccades are known to be modulated by perceptual attention⁵⁴, background⁵⁵, and salience⁵⁶.

While significant, the reduction in eye movements by fixation cross choice was subtle in effect size, and overall the distribution of saccades and microsaccades was similar irrespective of the fixation cross chosen. This highlights the need for additional measures to reduce the occurrence of eye movements in future studies.

5.3 The impact of fixation cross type on MVPA of visual information from eye tracking and EEG data

Our results are broadly consistent with previous studies that reported successful classification of diverse conditions of interest from eye movement alone, such as object category⁸ and grating orientation¹³ during perception, as well as stimulus information during the delay period of a working memory match to sample task⁹ or during the perception and attention phase of a working memory match to sample task¹⁰. We go beyond those studies by assessing the differential effect of fixation cross type on multivariate analysis. We observed that in multivariate pattern analysis on both eye tracking and EEG data accuracy was reduced for the bullseye fixation cross compared to the standard fixation cross. However, this effect was limited to object identity, and did not extend to object animacy. One reason for this might be that in our data set single object images were associated systematically with different eye movements that supported classification, whereas object image sets at the supra-category level of animacy were not and eye movements averaged out into similar distributions for animate vs. inanimate objects. However, this is likely a function of the stimulus set used here. In an experimental setup with two categories that have low intra-group image variability at the pixel level (e.g., highly controlled images of front-view faces and houses), we would expect systematic eye movement effects at the level of category.

The parallel reduction of classification accuracy for the bullseye fixation cross in the classification analyses based on eye tracking and on EEG data suggests a relationship between those observations. We substantiated this hypothesis by establishing that for the standard fixation cross, but not for the bullseye fixation cross, EEG and eye tracking were systematically related at the object image level. This is consistent with a previous study⁸ using a bullseye fixation cross that also observed both classification of visual stimulus information

from eye tracking and MEG data, but did not find a systematic relationship between them. We thus recommend the use of the bullseye fixation cross as a measure to reduce the confounding effect of eye movements in M/EEG, in particular but not limited to multivariate classification studies.

To put the effect into context, the identified confounding effect of eye movements on classification analysis was significant, but the effect size comparing the two fixation crosses (3.41%) was small compared to the overall classification results (26.22% and 23.98%, Fig. 4a). Together with the observation that the systematic relationship between EEG and eye tracking data was small or absent depending on the fixation cross, (Fig. 5b), this speaks for a limited confounding impact of eye movements systematically related to experimental conditions on EEG classification analyses under well-controlled laboratory conditions involving fixation control.

One open question is how exactly eye movements systematically related to object images influenced the EEG data and thus the classification results. One possibility is that the EEG acquisition picked up electrical activity created by eye movements. Another possibility is that the eye movements in effect led to changes in neural processing picked up by the EEG, e.g. elicited by changes in the visual input. Future studies relating eye movements to predicted changes in visual input combined with electromyography to isolate the effect of eye muscle activity are needed.

5.4 Limitations

Our results and conclusions are subject to several limitations. First, we measured data only from one eye, limiting the ability to distinguish measurement noise from microsaccades^{14,57}. Future studies recording both eyes are needed for further scrutiny. Second, it is unclear how well our results generalize from the particular experimental setup of centrally presented naturalistic images. Moving images, images presented in the periphery, or images presented alongside with other sensory cues⁵⁸ might affect other eye movement patterns that might be differently affected by the choice of a fixation symbol. Further studies assessing the effects of eye movements and their interaction with fixation crosses on M/EEG data are needed. Our data guide this research with the a priori hypothesis that the bullseye fixation cross will reduce eye movements compared to the standard fixation cross.

5.5 Conclusion

We conclude that the bullseye fixation cross reduced eye movements and their effects on M/EEG data for a visual setup with centrally presented naturalistic images. While the effect of eye movements on M/EEG data observed here was limited, and the reduction of eye movements and their associated effects on M/EEG is subtle, we recommend its use when tight control of eye movements is key for the experimental design and tested hypothesis.

6 Data availability

The code used for this project can be found under XX. The data set can be found on XXX.

7 Author contributions

G. H. and R. M. C. designed research; G. H. and A. P. C. acquired data; G. H. and A. P. C. analyzed data; G. H., and R. M. C. interpreted results; G. H. prepared figures; G. H. drafted manuscript; G. H., A. P. C., and R. M. C. edited and revised the manuscript.

8 Competing Interests

The authors declare no competing interests.

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